

01

Stellantis Line

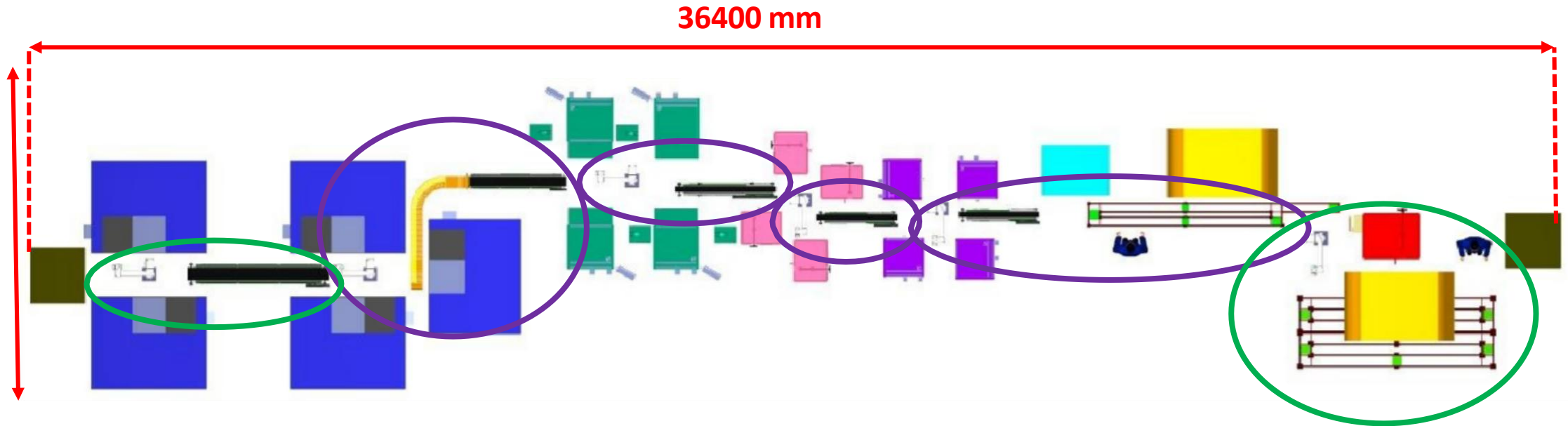
3D Isometric Layout



Complete Stellantis Sprocket Automation Line — Turning to Packing

Pick and Place Automation Topview

Part No. 1018



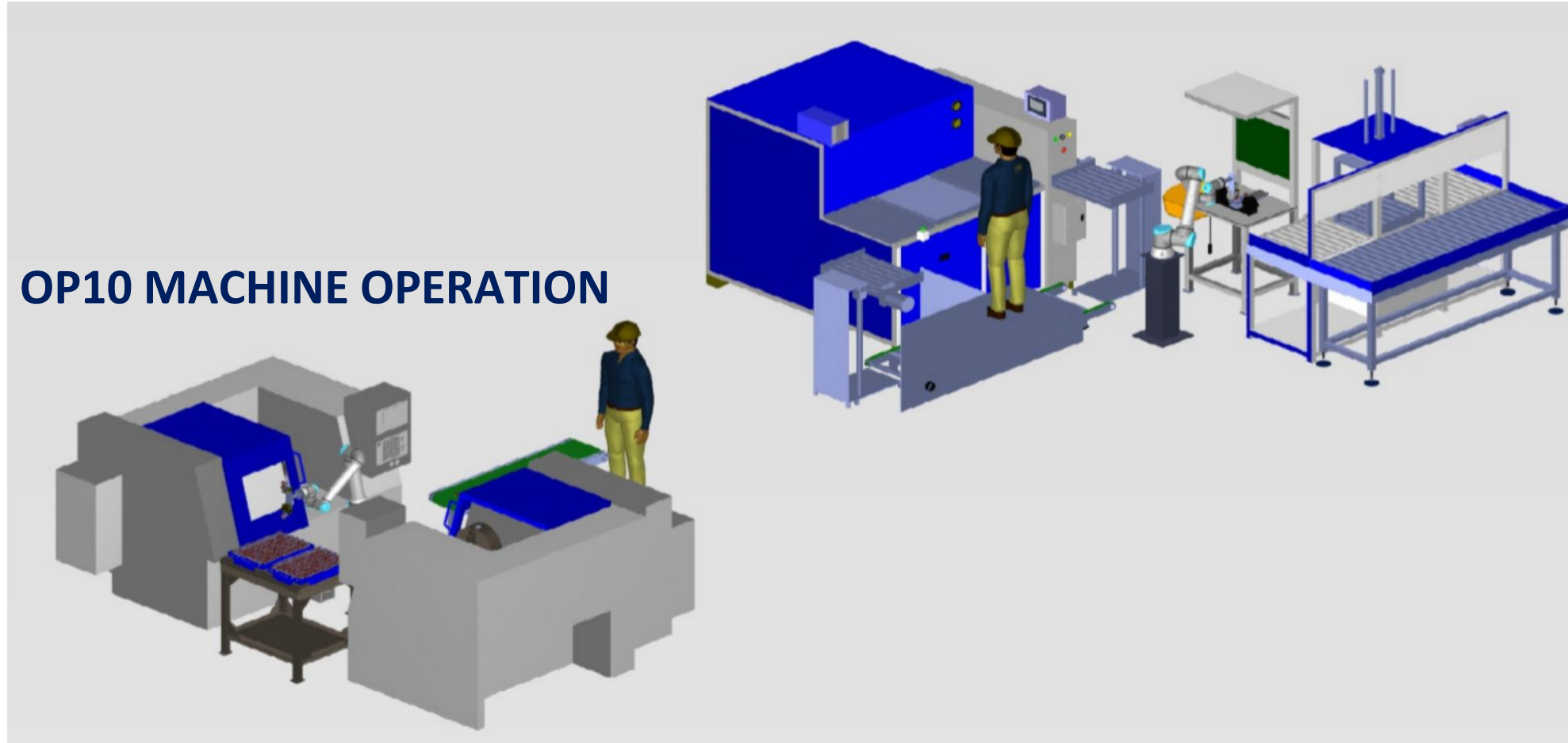
Phase 1 – Encircled with 

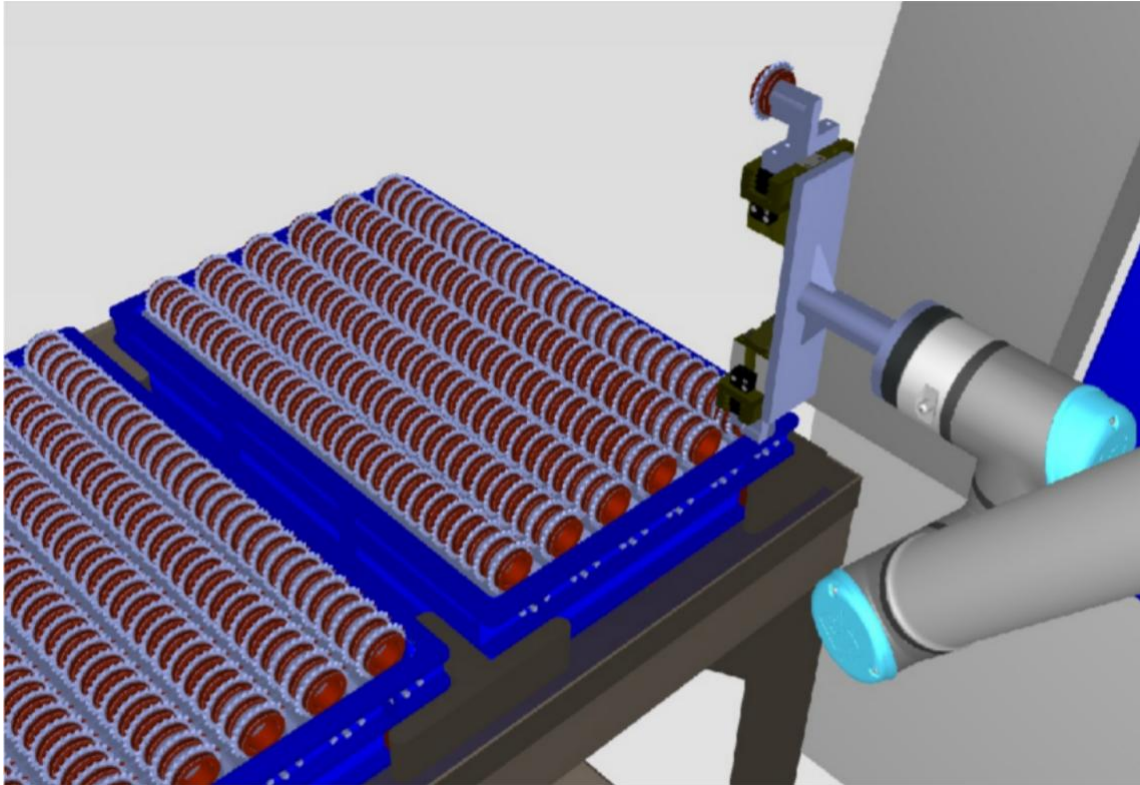
Phase 2 – Encircled with 

- **1 Phase Layout Details**

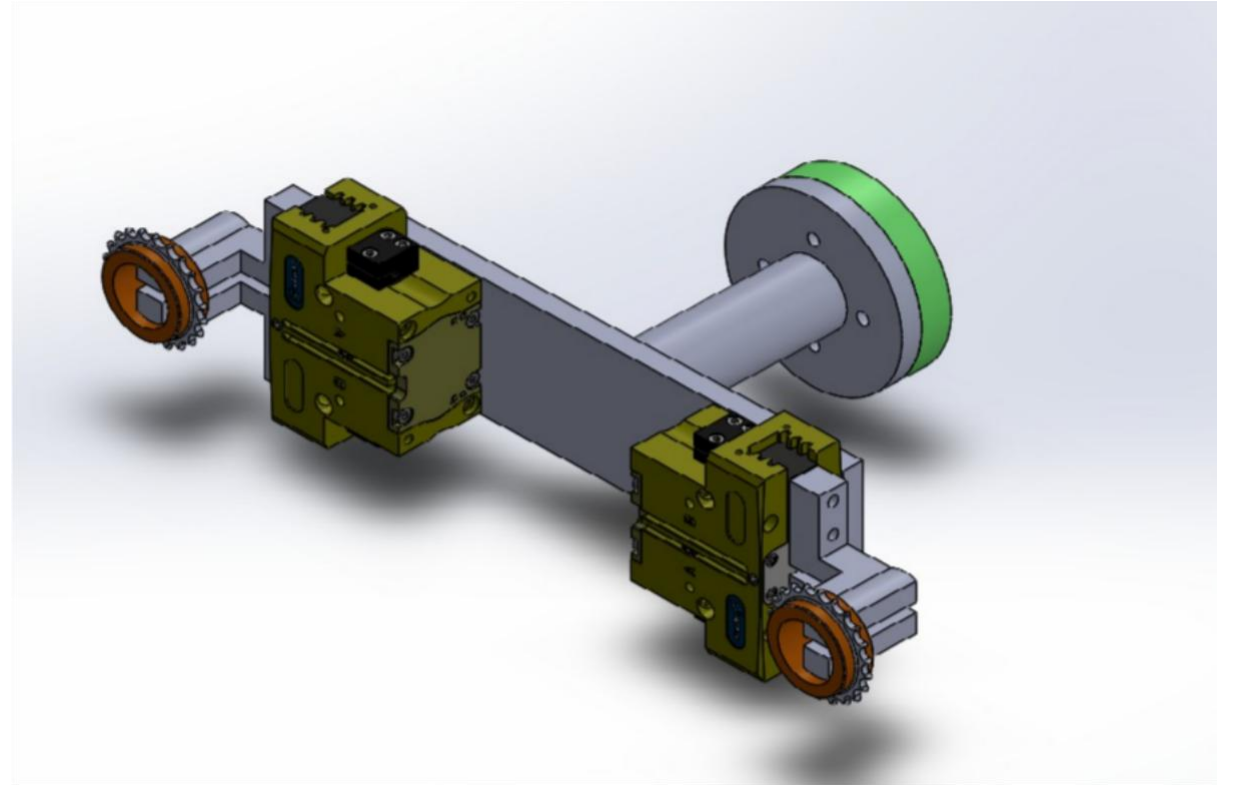
ULTRASONIC & INSPECTION MACHINE OPERATION

OP10 MACHINE OPERATION





OP10 INPUT TRAY



OP10 DUALGRIPPER

STELLANTIS LINE 1 - PHASE

- Ultrasonic & Inspection Scanner & Oil Wash Machine



- 1 Ultrasonic Wash
- 2 UR 12e
- 3 Inspection Scanner Unit
- 4 Oil Wash Unit

STELLANTIS LINE 1 - PHASE DAP APPROVAL (02-Apr-2026)

ROBONETICS
AUTOMATION SOLUTIONS LLP

Project Overview

FIELD	DETAILS
CUSTOMER NAME	L.G. Balakrishnan & Bros
PROJECT TITLE	Pick and Place Sprocket Line Automation
REVISION NO	Rev 00
DATE	20-03-2026
PREPARED BY	Sakthivel K – Asst. Project Manager
CHECKED BY	Karthikeyan S – Asst. Project Manager
APPROVED BY	Mr. Chandrasekar – Project Director
CUSTOMER APPROVAL & DATE	

JAYSEL SELVAN K *SINGARANI B* *Mr. Chandrasekar*
02/04/26 *02/04/26* *02/04/26*

REMARKS:

- 1) OP-10 operation DAP completed.
- 2) UW & vision inspection pick and place pending in DAP
- 3) UW Bin design need to check & scanning fixture need to check.

*Anand & attach all disc
Parts with this -
G. M. S.
02/04/26.*

STELLANTIS LINE 1 - PHASE (05-MAY-2026)

**MECHANICAL ITEAM ARE UNDER MANAFACTURING
INPROGRESS**



OP10 Output conveyar



UW conveyar



Cobot pedestal



OP10 input table

COBOT DELIVAY AT SUPPLIYAR END



STELLANTIS LINE 1 - PHASE (30-MAY-2026)

**MECHANICAL ITEAM ARE MANAFACTURING
COMPLETED**



OP10 Output conveyar



UW conveyar



Cobot pedestal



FAT – Major Pin Points / Observations

- **Edge Belt Conveyor**

Side guard installation to be completed.
Edge plate height to be reduced.

- **Bed Conveyor**

Guard fixing to be completed.
Belt misalignment and tracking issues identified; correction required.

- **Lifting Unit**

Roller bearing arrangement to be upgraded due to wobbling issue.
Taper roller bearing arrangement recommended for improved performance.
End caps and profile beading installation to be completed.

- **Roller Conveyor**

Noise issue identified and corrected.

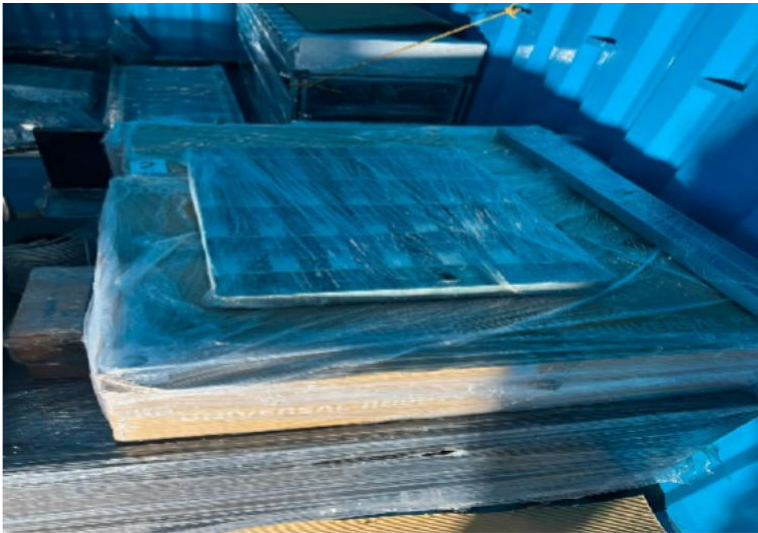
- **Controls & Robot Programming**

Cobot and PLC synchronization completed.
Minor changes identified during FAT and corrected.

- **Current Status:**

Major FAT points have been communicated to the integrator.
Required modifications have been completed.
Material is planned for dispatch along with the cobot today.

STELLANTIS LINE 1 - PHASE COBOT DELIVARY (20-JUN-2026)



**The Cobot, conveyor, pedestal, control panel,
and input table have been received at our end**

STELLANTIS LINE 1 - PHASE (27-JUN-2026)



OP10 Machine assemble as per the automation layout

STELLANTIS LINE 1 - PHASE



OP10 PRODUCTION VIDEO

STELLANTIS LINE 1 - PHASE

OP10 PROJECT – CNC1 & CNC2 PART COUNT REPORT

Period: 14/08/2026 to 23/08/2026

Operation: OP10

Shift: 3 Shifts

Date	DAY	Shift	CNC1 Part Count	CNC2 Part Count	Total Part Count	Target Count	Actual count	Remarks
14/08/2026	FRIDAY	1st	688	686	1374	3500	3709	
		2nd	731	709	1440			
		3rd	451	444	895			
15/08/2026	SATURDAY	HOLIDAY						
16/08/2026	SUNDAY	1st	596	575	1171	3500	1171	Sunday 1 shift Running
		2nd	0	0	0			
		3rd	0	0	0			
17/08/2026	MONDAY	1st	627	625	1252	3500	3311	Finger Issue & Door Issue
		2nd	492	470	962			
		3rd	562	535	1097			
18/08/2026	TUESDAY	1st	667	677	1344	3500	3730	
		2nd	659	654	1313			
		3rd	537	536	1073			
19/08/2026	WEDNESDAY	1st	701	627	1328	3500	3664	
		2nd	677	659	1336			
		3rd	500	500	1000			
20/08/2026	THURSDAY	1st	704	705	1409	3500	3217	3rd Shift CNC1 Not Running
		2nd	660	635	1295			
		3rd	0	513	513			
21/08/2026	FRIDAY	1st	634	626	1260	3500	3528	
		2nd	651	650	1301			
		3rd	487	480	967			
22/08/2026	SATURDAY	1st	694	591	1285	3500	2472	3rd Shift CNC1 & 2 Not Running
		2nd	591	596	1187			
		3rd	0	0	0			
23/08/2026	SUNDAY	1st	680	682	1362	3500	1362	Sunday 1 shift Running
		2nd	0	0	0			
		3rd	0	0	0			



Collaborative Cobot Operation

- While operating the collaborative cobot, one machine is running with the cobot.
- Simultaneously, another machine is being operated manually with a different component.
- This enables both automated and manual operations to run simultaneously.

STELLANTIS LINE 1 - PHASE (12-AUG-2026)

OP10 TRAINING AND HANDOVER

OPERATOR TRAINING RECORD			
Machine Tending Automation - UR12E Cobot with LMW CNC			
Project / Machines CNC Machine Tending System			
Customer Name	LGB		Place: Atanay
CNC Make: LMW	No. of CNC Machines: 2 Nos.		
Cobots: UR12E Cobot			
Material Loading: Input Table			
Finished Part Unloading: Output Conveyor			
1. TRAINING GIVEN			
The following training was provided to the Line In-charge regarding the operation and handling of the CNC machine tending automation system.			
S.No.	Training Topic	Training Given	Remarks
1	System Introduction	Explained the complete machine set-up system and working sequence.	Completed
2	UR12E Cobot	Explained robot operation, basic controls and operating procedures.	Completed
3	Input Table	Explained how material loading and unloading part placement on the input table.	Completed
4	CNC-I Operation	Explained CNC-I loading, machining cycle and unloading sequence.	Completed
5	CNC-II Operation	Explained CNC-II loading, machining cycle and unloading sequence.	Completed
6	Output Pick & Place	Demonstrated part picking from input table and placing into CNC machines.	Completed
7	CNC Unloading	Demonstrated removal of completed parts from CNC-I and CNC-II.	Completed
8	Output Conveyor	Explained finished part placement on the output conveyor.	Completed
9	Alarm Mgmt	Explained machine alarm, sequence monitoring and manual operation.	Completed
10	Manual Mode	Explained manual operation for checking and maintenance purposes.	Completed
11	Cobot Start / Stop	Explained safe procedure for starting and stopping the cobot.	Completed
12	Emergency Stop	Explained emergency stop locations and procedure to recover safely.	Completed
13	Safety Precautions	Explained safety precautions, restricted areas and safe operating practices.	Completed
14	Alarm Handling	Explained basic alarm, CNC and system alarm indications.	Completed
15	Troubleshooting	Explained basic checks for part pick failure, CNC cycle failure and conveyor issues.	Completed
16	Daily Checks	Explained checking of air pressure, material availability, gripper condition and machine status.	Completed
17	Shift Handover	Explained the procedure for communicating machine status and issues during shift handover.	Completed
18	Shutdown Procedure	Explained safe shutdown procedure after completion of production.	Completed

2. PRACTICAL TRAINING		Remarks
Practical demonstration and hands-on training were provided for:		Completed
Loading raw parts in the input table		Completed
Checking correct part orientation		Completed
Starting the UR12E robot		Completed
Operating the two LMW CNC machines		Completed
Cobot loading and unloading sequence		Completed
Transfer of finished parts to the output conveyor		Completed
Monitoring automatic cycle operation		Completed
Handling alarm and stoppages		Completed
Emergency stop operation		Completed
Safe recovery after machine/cobot stoppage		Completed
Final shift machine shutdown		Completed
3. SAFETY INSTRUCTIONS		Remarks
The Line In-charge was instructed to:		Completed
1. Ensure that all safety guards and safety devices are in proper condition.		Completed
2. Do not enter the robot operating area while the robot is in automatic operation.		Completed
3. Do not place hands inside the CNC machine during automatic operation.		Completed
4. Ensure proper part loading and orientation before starting the cycle.		Completed
5. Do not bypass or modify any safety device.		Completed
6. Press emergency stop in case of any unsafe condition.		Completed
7. Inform the maintenance/automation team in case of repeated alarm or abnormal operation.		Completed
8. Ensure adequate air pressure and proper gripper operation before production.		Completed
9. Keep the input table and output conveyor area clear.		Completed
10. Follow the approved operating sequence during production.		Completed
4. TRAINING ACKNOWLEDGEMENT		
Training on the above UR12E Cobot based CNC machine tending system with two LMW CNC machines, Input table and output conveyor has been completed.		
The Line in charge has understood the operating procedure, safety precautions and basic troubleshooting procedure.		
TRAINING GIVEN BY		
Name: K. SAKTHIARAJ & A. KRISHNA		
Designation: Asst. Manager / Staff Engineer		
Signature: [Signature]		
Date: 12/08/26		
TRAINING RECEIVED BY - LINE IN-CHARGE		
Name: TANILAKASU, RAMKUMAR, G. RAJESH R. Anandakumar, Sanyal		
Designation: Line In-charge		
Signature: [Signature], [Signature], [Signature], [Signature]		
Date: 12/08/26		
Total Person Attention - 6 individuals.		

STELLANTIS LINE 1 - PHASE OP10 ISSUE TRACKER

S.no	Query raised date	Pending work	Responsibility	Target date	Actual completion date	Status	Action to be taken / Follow
Pending work in OP 10							
1	15-07-2026	OP10 planger mechanism	Robonetics	28-07-2026		Not completed	
2	15-07-2026	Op 10 air planger	Robonetics	28-07-2026	01-08-2026	Completed	Installed dual air planger in op10
3	15-07-2026	OP10 input table	Robonetics	27-07-2026	20-08-2026	Completed	
4	27-07-2026	OP10 conveyar bur removal brash	Robonetics	27-07-2026	20-08-2026	Completed	
5	10-08-2026	OP10 conveyar L plate	LGB gear maintenance	01-09-2026		Completed	
6	27-07-2026	OP10 part present sensor	Robonetics	08-08-2026	20-08-2026	Completed	

STELLANTIS LINE 1 - PHASE - OP10

STELLANTIS LINE 1 - PHASE INSTALLATION TRACKER - OP10

S.No	Activity	Responsible	Start Date	Target Date	Duration	Status	Remarks
1	Material Receipt at Customer Site	RNT + Customer	22-06-2026	23-06-2026	2	CLOSED	Verify delivered materials
2	Unloading & Inspection	RNT	23-06-2026	23-06-2026	1	CLOSED	Check for transit damage
3	Machine Positioning & Alignment	RNT Mechanical Team	24-06-2026	25-06-2026	2	CLOSED	As per approved layout
4	Conveyor and Lifter Installation & Leveling	RNT Mechanical Team	25-06-2026	25-06-2026	1	CLOSED	Alignment and leveling
5	Cobot Pedestal Installation	RNT Mechanical Team	25-06-2026	25-06-2026	1	CLOSED	Anchor and level
6	Cobot Installation	RNT Automation Team	26-06-2026	26-06-2026	1	CLOSED	Robot mounting
7	Gripper Installation	RNT Automation Team	26-06-2026	26-06-2026	1	CLOSED	Mechanical & pneumatic connections
8	Pneumatic Line Installation	RNT Mechanical Team	26-06-2026	26-06-2026	1	CLOSED	FRL, tubing, valves
9	Control Panel Installation	RNT Electrical Team	27-06-2026	01-07-2026	4	CLOSED	Panel mounting
10	Electrical Wiring & Cable Dressing	RNT Electrical Team	27-06-2026	01-07-2026	4	CLOSED	Power and I/O wiring
11	PLC, HMI & Network Connections	RNT Automation Team	02-07-2026	06-07-2026	4	CLOSED	Communication setup
12	Power ON & I/O Verification	RNT Automation Team	07-07-2026	08-07-2026	2	CLOSED	Sensor and actuator checks
13	Robot Calibration	RNT Robot Engineer	09-07-2026	09-07-2026	1	CLOSED	TCP and frame calibration
14	PLC & Robot Program	RNT Automation Team	10-07-2026	11-07-2026	2	CLOSED	
15	Dry Run Trial	RNT Automation Team	13-07-2026	15-07-2026	3	CLOSED	Without production parts
16	Production Trial	RNT + Customer	16-07-2026	18-07-2026	3	CLOSED	Actual component validation
17	Cycle Time Optimization	RNT Automation Team	20-07-2026	21-07-2026	2	CLOSED	Performance tuning
18	Safety Validation	RNT + Customer	22-07-2026	22-07-2026	1	CLOSED	Safety interlocks and E-stop testing
19	Operator & Maintenance Training	RNT	23-07-2026	23-07-2026	1	CLOSED	Customer training
20	Site Acceptance Test (SAT) & Handover	RNT + Customer	31-07-2026	15-08-2026	2	CLOSED	Final approval

STELLANTIS LINE 1 - PHASE UW



UW Placed as per the automation layout

STELLANTIS LINE 1 - PHASE - UW

1018-part Stellantis - Ultra wash line Cobot Activity chart											Last updated on : 08.09.2026		
Sl.no	Activity	Responsibility	Plan vs actual	July'26				Aug'26				Status	Remarks
				I week	11 week	111 week	IV week	I week	11 week	111 week	IV week		
1	ultra wash machine position layout changed	Maintenance team	P			18.07.2026						Completed	Layout plan Only completed & waiting for Capex approval for further modification
			A			18.07.2026							
2	Ultra wash conveyer length & width - Modified	RNT & customer	P			21.07.2026						Completed	
			A			21.07.2026							
3	Compressor placement change over	Electric maintenance	P			21.07.2026						Under process	
			A										
4	Ultra wash bin- material modified from metal to hylam	RNT & customer	P				28.07.2026					Completed	
			A						27.08.2026				
5	Bin lock- modification	RNT	P				31.07.2026					Completed	
			A						27.08.2026				
6	Bin pushing- plate modification	RNT Mechanical team	P				31.07.2026					Under process	Gripper - means part holding under manufacturing process stage
			A										
7	keyence - rejection bin (flipper is optional)	RNT Mechanical team	P				24.07.2026					order in progress	order in progress
			A										
8	Cobot installation	RNT Mechanical team	P						24.08.2026			Completed	
			A						21.08.2026				
9	Programming for movement position	RNT Mechanical team	P						24.08.2026			Completed	Now only waiting for Gripper & production start up
			A						25.08.2026	05.09.2026			
			Plan		Actual		Under Progress			Delayed			

STELLANTIS LINE 1 - PHASE

STELLANTIS LINE 1 - PHASE TRACKER				
S No.	Task Description	Planned Date	Actual Date	Status
1	Requirement Date & Initial Feasibility	30-Jun-25	30-Jun-25	Closed
2	Plant Input & Cycle Time Study	03-Jul-25	10-Jul-25	Closed
3	Budgetary Estimation & RFQ	25-Jul-25	20-Aug-25	Closed
4	Vendor Pre-Qualification & NDA	09-Aug-25	04-Sep-25	Closed
5	RFQ Submission & Tech Discussion	11-Aug-25	11-Sep-25	Closed
6	Vendor Site Evaluation	10-Sep-25	05-Oct-25	Closed
7	Offer Evaluation & Tech Compliance	15-Sep-25	18-Oct-25	Closed
8	Risk Assessment & Commercial Negotiation	02-Oct-25	06-Dec-25	Closed
9	Techno-Commercial Comparison	08-Oct-25	30-Jan-26	Closed
10	Vendor Finalization & Mgmt Approval	12-Oct-25	16-Feb-26	Closed
11	Scope Freeze & PO Release	02-Nov-25	21-Mar-26	Closed
12	Advance Payment & Design Kick-off	30-Mar-26	03-Apr-26	Closed
13	Design Approval (Robotic + Gripper + Conveyor)	15-Mar-26	04-Apr-26	Closed
14	Cobot Delivery	17-Mar-26	15-May-26	Closed
15	Mechanical Fabrication (All Sub-systems)	20-Mar-26	20-May-26	Closed
16	Electrical / Controls Build	30-Mar-26	20-May-26	Closed
17	FAT	25-May-26	05-Jun-26	Closed
18	Dispatch	07-Jun-26	19-Jun-26	Closed
19	Installation & Commissioning	13-Jun-26		Open

PROJECT BOM - STELLANTIS LINE 1 PHASE

BOM	UOM	QTY	UNIT COST	TOTAL COST
Design & Engineering				
PLC Program	Lot	1	100000	100000
Project Management	Lot	1	50000	50000
Drawing and Documentation	Lot	1	30000	30000
Robots				
Collaborative Robots	Nos	2	1200000	1200000
Gripper				
Robots	Nos	7	100000	700000
Packing	Nos	3	150000	450000
Mechanical Accessories				
Pedestal	Nos	7	20000	140000
Belt conveyor	Lot	1	800000	800000
E&I				
HMI	Lot	1	200000	200000
Control Panel	Lot	1	500000	500000
Teach Pendant	Lot	1	200000	200000
Installation & Commissioning	Lot	1	300000	300000
Digitalization	Lot	1	300000	300000
TOTAL				₹62,00,000

NO OF MANPOWER REDUCED - STELLANTIS LINE 1 PHASE

Manpower Reduction -- Stellantis Line 1 Phase Automation						
S.No	Station / Operation	Manpower / Per shift (Before)	Manpower / Per shift (After Automation)	Manpower Reduced / Per shift	Avg. Monthly Cost per Head (₹) / Per shift	Monthly Cost Savings (₹) / Per shift
1	OP10 Machine Operation	2	1	1	₹22,000 x 1	₹22,000
2	Ultrasonic Wash, Vision & Oil Washing Station	3	1	2	₹22,000 x 2	₹44,000
TOTAL MANPOWER PER SHIFT		5	2	3		₹66,000

Summary	
Total Manpower Before (Manual)/ Per shift	5
Total Manpower After (Automation) / Per shift	2
Total Manpower Reduced / Per shift	3
% Manpower Reduction / Per shift	60.0%
Total Monthly Cost Savings / Per shift	₹66,000
Total Annual Cost Savings / Per shift	₹7,92,000

ROI CALCULATION - STELLANTIS LINE 1 PHASE

Parameter	Calculation	Value
Total Machines Automated		5
Total Cobots Used		2
Operators Required per Shift (Current)		5
Total Shifts per Day		3
Total Operators Required per Day (Current)	5×3	15
Salary per Operator per Month		₹ 22,000
Current Monthly Manpower Cost	$15 \times 22,000$	₹ 3,30,000
Current Annual Manpower Cost	$3,30,000 \times 12$	₹ 39,60,000
Operators Required After Automation		6
Monthly Manpower Cost After Automation	$6 \times 22,000$	₹ 1,32,000
Annual Manpower Cost After Automation	$1,32,000 \times 12$	₹ 15,84,000
Annual Savings Due to Automation	$39,60,000 - 15,84,000$	₹ 23,76,000
Total Automation Project Cost		₹ 62,00,000
ROI Payback Period (Years)	$62,00,000 \div 23,76,000$	2.6 Years
ROI Payback Period (Months)	2.6×12	31 Months
Operator Reduction	$05 - 02$	03 Operators Reduced
Savings After 5 Years	$(23,76,000 \times 5) - 62,00,000$	₹ 56,80,000

PO DETAILS - STELLANTIS LINE 1 PHASE

INVOICING PLAN – STELLANTIS ROBOTIC LINE PROJECT								
S No.	Milestone Payment %	Milestone Payment Description	Amount (₹)	Planned Invoice Date	Actual Invoice Date	Invoice No.	Payment Status	Remark
1	70%	Advance to Supplier	₹43,40,000	Mar-2026 (PO date)	30-Mar-2026		Paid	Advance paid
2	20%	Upon Delivery at Site	₹12,40,000	Jun-2026	19-Jun-2026		Paid	Delivery at Site
3	10%	Upon Installation & Commissioning	₹6,20,000	Jul-2026			Pending	Installation & Commissioning
Total	100%		₹62,00,000					